

Potential Benefits of Lean Integration

Lean presents several compelling advantages as a backend verification engine for an educational natural deduction system.

First, Lean provides **machine-verified correctness guarantees** grounded in dependent type theory. Unlike a handcrafted rule engine, Lean's kernel ensures that any accepted proof satisfies strict logical consistency. Integrating Lean would elevate the system from structural validation to formally verified proof checking, eliminating the possibility of subtle implementation errors in the local rule engine.

Second, Lean has developed a rapidly expanding ecosystem, particularly through the **mathlib** community project. This library contains extensive formalised mathematics and logical foundations, making Lean not only suitable for propositional logic but also scalable toward predicate logic, type theory, and more advanced formal systems. Such scalability aligns well with long-term academic use.

Third, Lean's increasing adoption in undergraduate and postgraduate teaching makes it pedagogically relevant. Many universities now introduce Lean in logic, formal methods, and theorem proving courses. Integrating Lean could therefore:

- Familiarise students with industry-relevant formal verification tools
- Provide continuity between educational exercises and research-level tools
- Bridge the gap between classroom logic and real-world formal verification

Another advantage is Lean's modern tooling infrastructure. Lean supports:

- Interactive feedback via VS Code extensions
- Web-based editors
- Script-based proof representation

This tooling makes it feasible to design a hybrid architecture in which user-generated proof steps are translated into Lean syntax and verified server-side.

Finally, Lean would allow the system to move beyond propositional natural deduction. While the implemented engine focuses on propositional logic, Lean supports:

- First-order logic
- Higher-order logic
- Dependent type systems

Thus, Lean integration would significantly enhance the formal robustness, scalability, and academic credibility of the system.